2026 Q1 Gaming Laptops Market API (1.0.0)

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A RESTful API for querying and managing gaming laptop market data. Built with FastAPI and SQLite. Features Basic Authentication for modifying data.

Laptops

Get Laptops

QUERY PARAMETERS

`limit`	integer (Limit) Default: `10`
`skip`	integer (Skip) Default: `0`

Responses

> **200** Successful Response

> **422** Validation Error

GET `/laptops`

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
        "title": "string",
        "brand": "string",
        "price": 0,
        "discount_pct": 0,
        "price_currency": "$",
        "stars": 0,
        "reviews_count": 0,
        "id": 0
    }
]
```

Create Laptop

AUTHORIZATIONS: > *HTTPBasic*

REQUEST BODY SCHEMA: application/json
required

`title` required	string (Title)
`brand` required	string (Brand)
`price` required	number (Price)
`discount_pct`	number (Discount Pct) Default: `0`
`price_currency`	string (Price Currency) Default: `"$"`
`stars`	number (Stars) Default: `0`
`reviews_count`	integer (Reviews Count) Default: `0`

Responses

> **201** Successful Response

> **422** Validation Error

POST /laptops

Request samples

Payload

Content type
application/json

Copy

```
{
    "title": "string",
    "brand": "string",
    "price": 0,
    "discount_pct": 0,
    "price_currency": "$",
    "stars": 0,
    "reviews_count": 0
}
```

Response samples

201 **422**

Content type
application/json

Copy

```
{
    "title": "string",
    "brand": "string",
    "price": 0,
    "discount_pct": 0,
```

```
    "price_currency": "$",
    "stars": 0,
    "reviews_count": 0,
    "id": 0
}
```

Get Laptop

PATH PARAMETERS

`laptop_id` required	integer (Laptop Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /laptops/{laptop_id}

Response samples

200 | 422

Content type
application/json

Copy


```
{
    "title": "string",
    "brand": "string",
    "price": 0,
    "discount_pct": 0,
    "price_currency": "$",
    "stars": 0,
```

```
    "reviews_count": 0,
    "id": 0
}
```


Update Laptop

AUTHORIZATIONS: > *HTTPBasic*

PATH PARAMETERS

`laptop_id` required	integer (Laptop Id)

REQUEST BODY SCHEMA: application/json
required

`title` required	string (Title)
`brand` required	string (Brand)
`price` required	number (Price)
`discount_pct`	number (Discount Pct) Default: `0`
`price_currency`	string (Price Currency) Default: `"$"`
`stars`	number (Stars) Default: `0`
`reviews_count`	integer (Reviews Count) Default: `0`

Responses

> **200** Successful Response

> **422** Validation Error

PUT /laptops/{laptop_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "title": "string",
    "brand": "string",
    "price": 0,
    "discount_pct": 0,
    "price_currency": "$",
    "stars": 0,
    "reviews_count": 0
}
```

Response samples

200 **422**

Content type
application/json

Copy

```
{
    "title": "string",
    "brand": "string",
    "price": 0,
    "discount_pct": 0,
    "price_currency": "$",
    "stars": 0,
    "reviews_count": 0,
    "id": 0
}
```

Delete Laptop

AUTHORIZATIONS: > *HTTPBasic*

PATH PARAMETERS

`laptop_id` required	integer (Laptop Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /laptops/{laptop_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { ... }
    ]
}
```

Advanced Analytics

Search Laptops

Advanced search endpoint demonstrating dynamic SQL WHERE clause generation. Safely utilizes parameterized queries to prevent SQL injection.

QUERY PARAMETERS

`min_price`	number (Min Price) Minimum price in USD
`max_price`	number (Max Price) Maximum price in USD
`min_stars`	number (Min Stars) Minimum star rating (0-5)
`brand`	string (Brand) Specific brand name (e.g., ASUS, MSI)

Responses

> **200** Successful Response

> **422** Validation Error

GET `/search`

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all


```
[
  - {
        "title": "string",
        "brand": "string",
        "price": 0,
        "discount_pct": 0,
        "price_currency": "$",
        "stars": 0,
        "reviews_count": 0,
        "id": 0
    }
]
```

Get Brand Analytics

Aggregation endpoint demonstrating GROUP BY, HAVING, AVG, COUNT, and ORDER BY. Calculates key market metrics for each laptop brand, with dynamic filtering.

QUERY PARAMETERS

`min_models`	integer (Min Models) Default: `1` Minimum number of laptop models the brand must have to be included
`limit`	integer (Limit) Default: `10` Limit the number of brands returned (e.g., Top 10 brands)

Responses

> **200** Successful Response

> **422** Validation Error

GET /analytics/brands

Response samples

200 422

Content type
application/json

Copy

```
null
```

Get Top Deals

Retrieves the laptops with the highest discount percentages. Demonstrates filtering out zero-discount items and sorting descending.

QUERY PARAMETERS

`limit`	integer (Limit) Default: 5 Number of top deals to return

Responses

> **200** Successful Response

> **422** Validation Error

GET /analytics/top-deals

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all


```
[
 - {
        "title": "string",
        "brand": "string",
        "price": 0,
        "discount_pct": 0,
        "price_currency": "$",
        "stars": 0,
        "reviews_count": 0,
        "id": 0
    }
]
```